Milestone 03

PDF edition of preserved repository document | docs/milestone_03.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

First learned Mindscape milestone

IMPLEMENTED

Small trainable NumPy backend; answer-only baseline (A); trajectory-supervised state/history-conditioned policy (B); environment-interactive inference; action masking and unmasked diagnostics; resettable working memory; partitioned SQLite episodic API; prefix/final verification; deterministic Adam training, validation checkpoint selection; save/load and CLI; shared evaluator integration; tracking, failures and preliminary curves.

TESTED

All original 54 tests plus 15 learning tests passed. Coverage includes numerical gradient checks, optimizer learning/reproducibility, regime adapters, serialization, valid masks, memory reset, illegal actions, timeout/step limits, trajectories, verifier integration, configuration, checkpoint reload, SQLite partitions and train-to-evaluation smoke runs.

VERIFIED

10 actual trained checkpoints were reloaded before evaluation; 30 learned evaluations used identical held-out splits. Checkpoint/dataset/budget/regime compatibility is checked. The trained unmasked policy executes 19×3 to 57 with five legal actions and a complete verified trajectory. The baseline checkpoint predicts 28 for this separate demonstration. Each selected action, actual event/result and before/after state is retained in examples/19_times_3_learned.json. Working memory resets and no episodic retrieval is used.

EXPERIMENTS RUN

50 examples: baseline and policy at seeds 0,1,2. 100 and 250 examples: each at seed 0. Each model evaluated on 500 IID and 500 OOD examples; policy also evaluated unmasked. Two additional untrained unmasked controls (seed 0) reached 0% IID and 0% OOD success.

One early policy-training development smoke run is retained separately under results/.

No hyperparameters or test sets were changed after the fixed comparison started.

ACTUAL RESULTS

All values below are measured preliminary development results, not final thesis findings.

At 50 examples: mean ± sample standard deviation across three seeds; other sizes: one seed.

Training examples	Model	IID exact accuracy	OOD exact accuracy
50	baseline	0.267% ± 0.306%	0.000% ± 0.000%
50	mindscape	100.000% ± 0.000%	100.000% ± 0.000%
50	mindscape_unmasked	100.000% ± 0.000%	82.000% ± 4.084%
100	baseline	0.400%	0.000%
100	mindscape	100.000%	100.000%
100	mindscape_unmasked	100.000%	76.400%
250	baseline	0.400%	0.000%
250	mindscape	100.000%	100.000%
250	mindscape_unmasked	100.000%	78.800%

Policy supplied-trajectory validity is 100% for both masked and unmasked runs: illegal proposals are rejected, so failed episodes retain valid incomplete prefixes. Groundedness and goal success equal each policy run’s final accuracy; unsupported rate is its complement. Baseline groundedness and goal success are 0%, unsupported rate 100%, and trajectory validity null (no supporting trace). All five trained policies achieved 100% teacher-forced unmasked validation action accuracy. Seed-0 policy validation loss at 50 examples fell from 1.409832 to 0.002929.

For the unmasked 50-example OOD policy: 106/67/97 invalid-action failures at seeds 0/1/2 respectively. Full taxonomy counts and every prediction are saved. No DER or architecture-superiority conclusion is drawn. Masked 100% is guaranteed by one legal action per state. The policy delegates exact arithmetic to the environment; the baseline learns answer digits without that tool. Even unmasked action success does not establish learned arithmetic or a fair data-efficiency advantage.

NOT IMPLEMENTED

Regime C feedback learning (interface only), pretrained language-model integration, PEFT/LoRA, memory-based learning/retrieval experiments, generated explanations, final statistical studies, matched-tool controls, multiple-choice environment redesign, UI, multimodality, dream engine and historical GPT/Gemini comparison.

KNOWN LIMITATIONS

Singleton masking and exact arithmetic access confound comparison. The baseline is very weak; failure is retained and cannot support general conclusions. More transition labels per policy problem and differing heads mean supervision/capacity are unmatched. Validation policy labels use correct histories (teacher forcing); OOD rollout errors expose different execution behavior. Development dataset is unbalanced and not a final untouched research set. Only 50-example points have three seeds. Training uses a fixed 400-update budget rather than equal epochs. The learned adapter is multiplication-only; core/benchmark remain general. Features support six-digit operands, eight-digit answers. TensorBoard and gradient-framework tooling are absent; plain JSON logs are used. Inference checkpoints do not preserve resume-training Adam state. Optional learning tests skip without NumPy. Floating point/RNG may differ across library/CPU versions. RAM could not be queried; PNG chart rendering is unavailable, so standard ReportLab SVG curves are delivered. Ruff/MyPy/pytest and hosted CI were not run.

MODEL USED

Local randomly initialized task-specific NumPy tanh MLP, not a foundation model. No downloaded model, API, pretrained weights or paid dependency. See docs/modeling.md.

MODEL SIZE

Shared backbone: 52 inputs, 64 hidden units, 3,392 parameters. Baseline: 8,722 total; policy: 3,652 total. Baseline eight decimal digit heads plus sign; policy four action logits.

DATASET SIZES

Development v1: train 1000, validation 250, IID test 500, OOD test 500.

Learning budgets 50/100/250 use identical recorded nested subset IDs for each model.

Policy row counts: 50 examples → 376 transitions, 100 examples → 740 transitions, 250 examples → 1783 transitions.

SEEDS

Dataset seed 42, fixed subsets. Training/mini-batch seeds 0,1,2 at budget 50;

seed 0 at budgets 100/250; untrained control seed 0.

TRAINING TIME

Ten fixed-comparison optimization runs total 0.553455 seconds.

Per-run seconds (optimization plus validation only):

Regime	Budget	Seed	Seconds
answer_only	50	0	0.076317
trajectory_supervised	50	0	0.035391
answer_only	50	1	0.074609
trajectory_supervised	50	1	0.035283
answer_only	50	2	0.074834
trajectory_supervised	50	2	0.035265
answer_only	100	0	0.077495
trajectory_supervised	100	0	0.034872
answer_only	250	0	0.074412
trajectory_supervised	250	0	0.034979

INFERENCE TIME

Thirty learned evaluations total 20.205644 seconds, including prediction/assessment.

Each run covers 500 examples. Full per-run times and calls remain in

experiments/first_learned/curves.json; model calls are 500 per baseline run and

3,944 IID / 6,635 OOD per complete policy run. Failed unmasked OOD runs use fewer calls.

TEST COUNT

69 passed on Python 3.12.14 / NumPy 2.3.5.

GIT STATUS

Branch main; prior history preserved. Final clean status checked after committing.

LATEST COMMIT

Training protocol committed before fitting: f33ce57; model implementation: ef3d94e.

This report and measured snapshots are included in the final milestone commit shown in chat.

REPOSITORY TREE

```
src/mindscape/models/{base,encoding,numpy_backend,learned}.py
src/mindscape/training/{adapters,optimizer,run}.py
src/mindscape/memory/{working,episodic}.py
src/mindscape/evaluation/{runner,evaluator,schemas}.py
configs/experiments/learning.toml
scripts/{train,evaluate,run_learned_episode,run_learning_experiment,plot_learning_curves}.
py
docs/{modeling,training,experimental_protocol,milestone_03}.md
examples/19_times_3_learned.json
experiments/first_learned/{curves,failure_analysis,fairness,untrained_controls}.json
experiments/first_learned/{test_accuracy,ood_test_accuracy}.svg
results/learned_development_v1/<model_budget_seed>/
  config.yaml, config.json, metadata.json, metrics.json, training.jsonl, checkpoint/
results/learned_development_v1/evaluations/<unique_id>/
  config.json, metrics.json, predictions.jsonl, summary.md
results/learned_development_v1/{comparison,curves,failure_analysis,19_times_3_learned}.jso
n
results/learned_development_v1/controls/<unique_id>/
tests/{test_foundation,test_benchmark,test_learning}.py
```